



WEAVE Open-Time Proposal

Title

Kinematics of nuclear and outer shells in elliptical galaxies

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Summary

We propose to confirm shell-like kinematics in elliptical galaxies using the high velocity resolution of the WEAVE LIFU. In several prototype shell galaxies, WHT/ISIS long-slit spectroscopy revealed spatially-narrow velocity features. They were tentatively interpreted as kinematic shells, or nuclear kinematic signatures of the accretion events that create photometric shells visible in the outer parts. With the 3D capabilities of the WEAVE LIFU we will obtain full 2D kinematic maps and confirm or disprove kinematic shells. We also aim to show WEAVE can measure l.o.s. velocities of the brightest outer shells in the sample to distinguish between radial and tangential orbits. With the large wavelength coverage in LR mode, we will explore stellar population signatures in the nuclear shells such as distinct line index differences, and see if the distribution of emission lines also traces shell morphologies.

Unveiling the build-up of massive elliptical galaxies remains an important goal nearly 50 years after Alar Toomre's 1977 seminal paper on the merger hypothesis. The problem has become increasingly interesting from work at low z (global scaling relations implying homogeneity; stellar populations showing old stellar ages) and at high redshift, eg., downsizing (Perez-Gonzalez+2008), up-sizing (Franx et al 2008); red-nuggets at high redshift (Damjanov+2011), and bulge-disk morphology at these redshifts (Costantin+2021). A favored emerging picture for massive ellipticals is the two-phase formation: an initial dissipational collapse possibly fed by cold flows (Dekel+2009) that creates the inner half of the mass at z 2 to 5, followed by an accretion phase $z \sim 1.5$ to 0, when the galaxy grows in size due to minor mergers, with size growth being much larger than the growth in mass and the increase of the central velocity dispersion (refs). Size growth driven by minor mergers since z 1.5 is supported by $z \sim 1$ velocity dispersions (Peralta de Arriba+2015) and merger-tree simulations (Frigo & Balcells 2017).

Shell galaxies are good laboratories to obtain clues about the nature of the second, accretion phase. Shells, discovered by Malin & Carter (1983), were soon explained from radial minor mergers through the process of phase wrapping (Quinn 1984). In this mechanism, accreted material goes through the centre at high speed and it does not settle there. In the nineties, Carter and collaborators used the exquisite spectral resolution of the then new ISIS spectrograph on the WHT to study the core kinematics of shell galaxies. They unveiled narrow velocity features in the nuclei of two prototype shell galaxies: NGC 7626 (Balcells & Carter 1993) and NGC 474 (Hau+ 1996). See Figs 1, 2.

Because these features are seen along several position angles, Balcells & Carter (1993) hypothesized that they might represent kinematic shells in the nuclear regions of galaxies with merger signatures such as shells and kinematically-decoupled cores (KDCs). If indeed linked to kinematic shells, these features would imply a high-density dynamical subsystem oscillating in the galaxy nucleus, quite unlike what's expected in phase-wrapped or spatially-wrapped shells, and, importantly, not immediately-expected in the two-phase model for the growth of massive ellipticals. In NGC 474 (see Alabi+ 2022), Balcells & Carter (unpublished) found strong suggestion of double-peaked l.o.s. velocity distributions which could however not be fully confirmed, despite a spectral resolution of $R=6000$ (see Fig. 3). Confirming double-peaked velocity distributions would open the door for a wide range of studies on the nature of the accretion process and the nature of the merger precursors, including stellar population diagnostics. The distribution of [OIII] is highly irregular, with a suggestion of a shell morphology to the North of the nucleus (Sarzi+2006). SAURON data (Emsellem+2004) also gives a suggestion of kinematic shells despite the low velocity resolution.

The radial/tangential nature of pre-merger orbits leading to shells can be constrained by measuring line-of-sight velocities of outer shells. This is a difficult measurement due to the low surface brightness of shells. Balcells et. al (2000) published velocity measurements for the bright outer shell of NGC 3656, using WHT/INTEGRAL spectroscopy, but the uncertainty of the measurement prevented a diagnostic of the shell being radial (phase wrapping) or with significant tangential velocity (spatial wrapping). The other measurement is for the outer shell of NGC 474, using the small-format IFU of Keck/KCWI (Alabi+ 2020).

The present proposal aims to obtain 3D spectroscopic information for four prototype shell galaxies with high-quality HST imaging and ISIS multi-long-slit spectroscopy, both from previous collaborations (Carter and others). Spectroscopic information is key to the interpretation of the structure that we see at high spatial resolution in the ACS images, and will be used:

1. To obtain high-velocity resolution full 2D stellar kinematic maps of the nuclei of four galaxies being candidates to have kinematic shells. 2D maps of V , σ and high-order moments h_3 and h_4 will uniquely constrain subcomponents, including kinematic shells, providing a constrain on the dynamical age of the cores. Kinematics will be derived from Mg, H-Alpha and NIR CaII regions, which will provide information on dust effects on the kinematics.
2. To obtain emission-line velocities for the gas being disrupted by the core of the remnant, and absorption-line velocities of the young stellar disks, to see whether the gas is being accreted in a disk, or as spherical accretion or inflow. Mergers will lead to extended gas distributions in the majority of remnants (Sarzi+ 2006), and the accretion of cold gas can lead to the direct formation of a young, kinematically decoupled component (e.g. NGC 5953: Falcon-Barroso+ 2006).
3. To obtain deep $R=2500$ spectroscopy to infer l.o.s. velocities of brightest outer shells in the sample. This will provide dynamical confirmation of whether the shells are phase-wrapped or space-wrapped, and opening the way for detailed merger modeling.
4. To exploit the wide wavelength coverage to extract stellar population parameters and identify peculiar stellar populations in the nuclei or associated to the kinematic shells.

For the nuclear kinematics, both $R=10,000$ (HR) and $R=2500$ (LR) data will be obtained. HR (Mg, and Halpha regions) will provide the resolving power required to map the kinematic substructures and any peculiar line-of-sight velocity distributions such as double-peaked distributions in NGC 474. LR will provide dynamical and line-index maps over a wide wavelength range. For some galaxies, a single LIFU pointing covers both the core and the outer shells.

The proposed observations will advance our knowledge on the nature of the merger orbits and the contribution of the accreted material to build up the high-density nuclei and the outer envelopes of ellipticals. Also, we will advance on the knowledge of new stellar populations originating in the merger event.

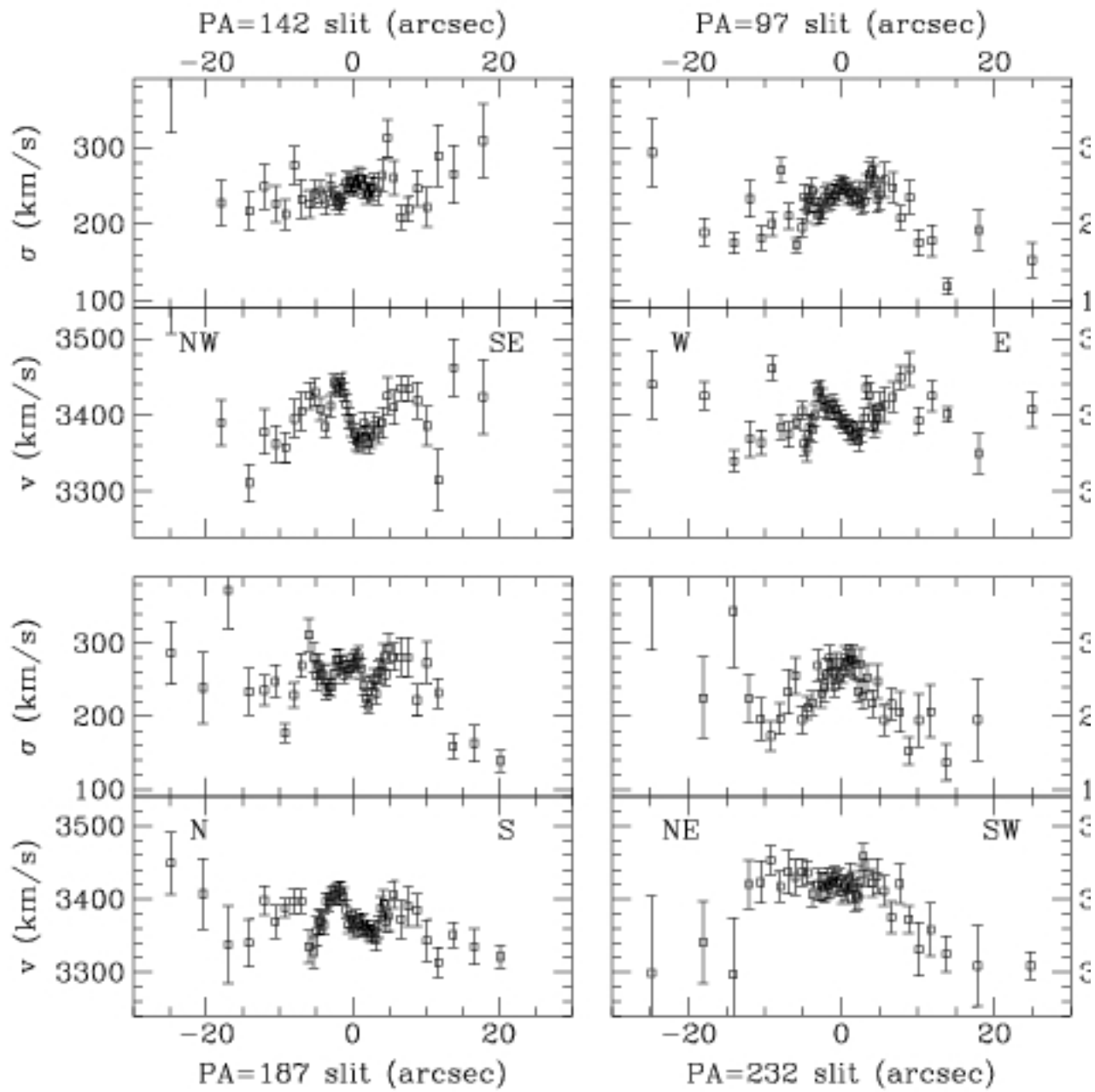


Figure 1: Four velocity and dispersion profiles for NGC 7626, from Balcells & Carter (1993). Sharp velocity features ('kinematic shells') are found along position angles 97, 142, 187.

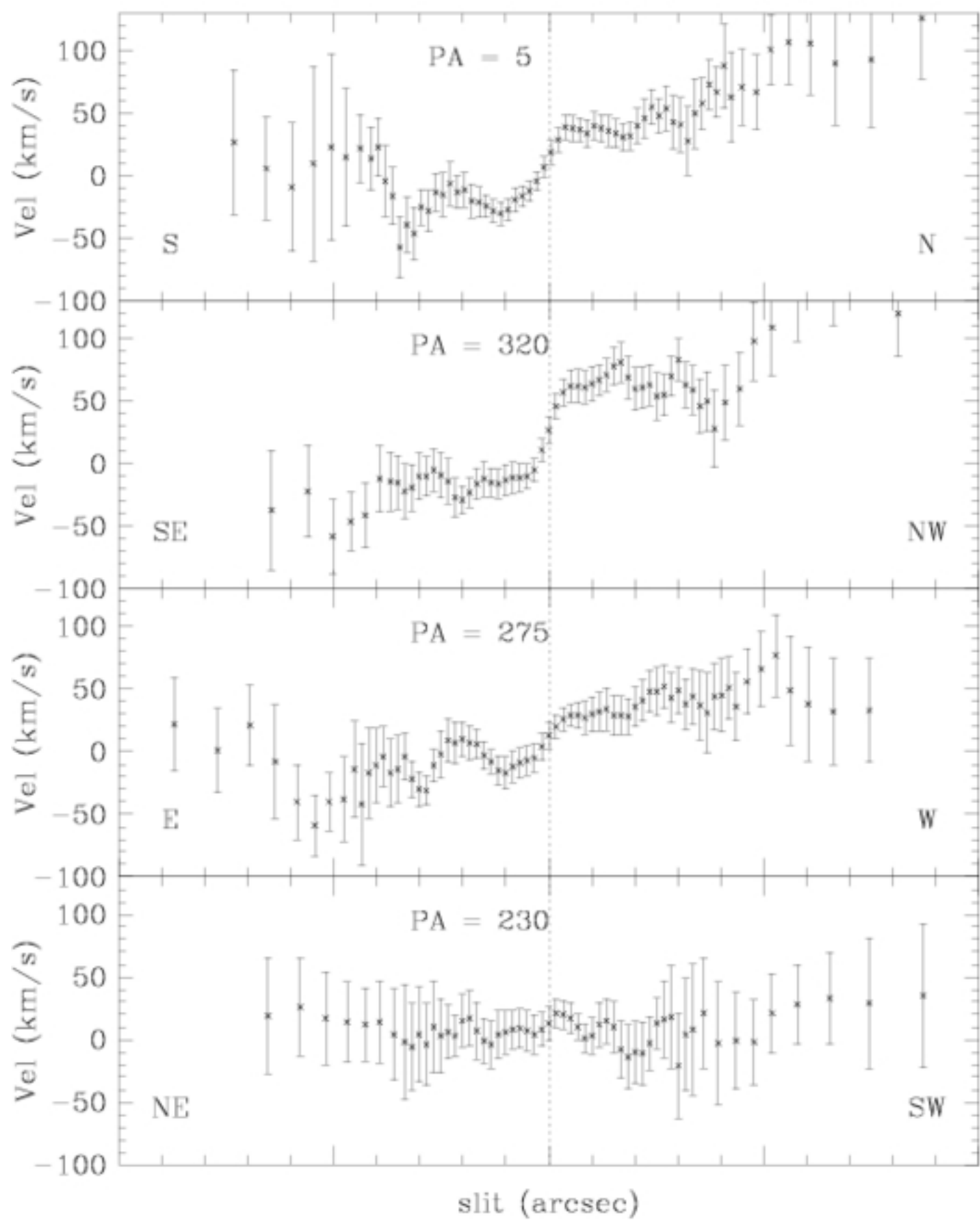


Figure 2: Four velocity profiles for NGC 474, from Hau et al (1996). Sharp velocity features are found along position angles 5 and 275.

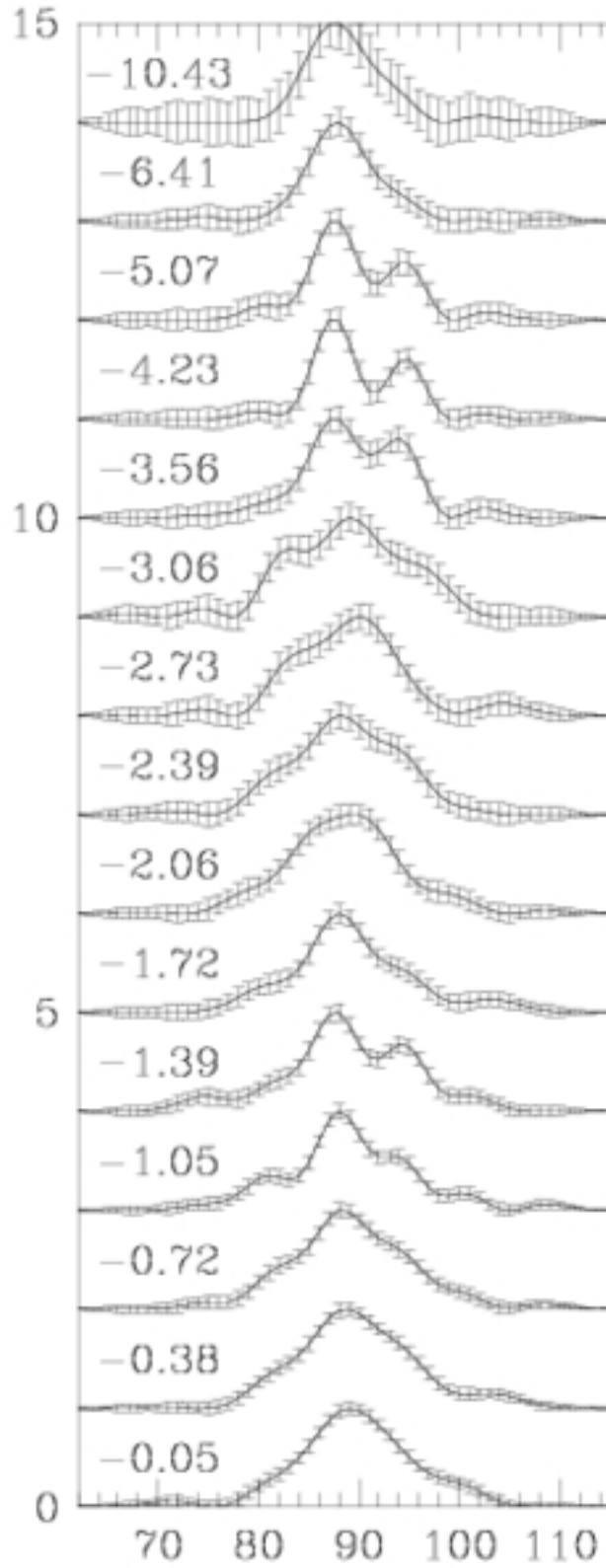


Figure 3: Line-of-sight velocity distributions (fv) for NGC 474. They correspond to positions from the center to -10.43" along position angle 275. A strong suggestion of double-peaked fv appears on 3 bins from 3.5" to 5.07". The fv were determined using unresolved gaussian decomposition (Kuijken & Merrifield 1993). The x-axis gives line-of-sight velocities in units of 25.81 km/s.

Technical case

Sample

The sample comprises four prototype ellipticals/S0 well studied by our group. We have previous HST/ACS imaging and WHT/ISIS spectroscopy. The four galaxies have shells and nuclear peculiar kinematics.

Observational strategy

The WEAVE LIFU is the instrument of choice for this project owing to its large FOV, its high throughput and its unique HR spectral resolution $R=10,000$. Its main draw-back is its coarse spatial sampling, which will smear the signal of narrow velocity features (typically 2 arcsec width). VLT/MUSE provides an better spatial sampling but is not a good alternative owing to its low $R=2000$ @ 0.46 micron. We will improve the spatial sampling by taking dither-6 observations: the depth requirements for outer shells need 2-hr exposures in any case. As noted below, spatial sharpness will be maximized by using cube reconstruction tools developed by our team.

For the HR observations, the blue arm will be fitted with the Green VPH, covering H-Beta, [OIII], Mg and Fe lines. This is an ideal region for both stellar and ionized-gas kinematics. The red VPH, centered on H-Alpha, will deliver exquisite emission-line profile information as well as sensitive star-formation indices and extinction estimates from H-Alpha / H-Beta ratios.

In the LR observations we want to obtain the vast array of population indices available in the full wavelength range of WEAVE in LR. Care will be exercised to ensure that flux standard observations will yield accurate flux calibration and telluric correction.

Exposure times are established by the SNR requirements inferred from our previous spectroscopy. We provide sample calculations for NGC 474, which has V-surface brightness of 19.5 mag/sq.as (radius of 8 arcsec, see Sikkema+2007). Our N474 longest exposure was $t_{\text{exp}}=9000$ s and gave $S/N=30$ per (spatial x spectral) pixel = (0.3sq.as x 26 km/s). With WEAVE HR2 (green), we bin x2 in velocity, yielding 12 km/s/pix which provides a nice x3 sampling of the spectral resolving element of 30 km/s. Thanks to the large spatial aperture of the LIFU, 5.3 sq.as, LIFU exposures deliver the same SNR as ISIS, per spectral pixel, with an exposure time 0.12 times that of ISIS. To obtain $S/N = 30$ per spectral bin of 0.2 Å/pix, we need $S/N=67$ per Angstrom. According to the WEAVE ETC (<https://astro.ing.iac.es/signalWEAVE/>), 2400 s are needed to deliver $S/N=67$ for $V=19.5$ mag/sq.as. We therefore need a standard dither-6 for HR observations.

Exposure times for LR are derived from the requirement to obtain velocities for outer shells. The outer shell of NGC 3656 has surface brightness 22 R-mag/sq.as (Balcels+2001), whereas that of NGC 474 has 23.9 r-mag/sq.as (both Vega). Using signalWEAVE.5.0, 2-h integration coadding 8 LIFU fibres on NGC474 yields S/N per Å = 17.5, and a radial velocity error of 10 km/s, adequate to constrain radial vs tangential shell motion.

Dark time is necessary given the low surface brightness of the outer shells, whereas good seeing is needed for the tracing of the inner kinematic shells. Seeing requirements for each OB will be specified in Phase 2.

Processing and analysis

Basic processing will use the standard CPS and APS pipelines, but additional tools will need to be used. Most important will be a cube reconstruction that maximizes spatial resolution, see Sanchez et al (2023), taking advantage of high-quality imaging of these well-studied galaxies.

Double-peaked velocity distributions will be investigated with unresolved-Gaussian decomposition (Kuijken & Merrifield 1993), and with non-parametric extraction methods such as BAYES-LOSVD (Falcon-Barroso & Martin 2021).

For the extraction of stellar and gas kinematics of outer shells, we will go beyond standard WEAVE APS stellar kinematic analysis, by using all available features, from [OII]3727 all the way to CaII-8550.

References:

Alabi et al, 2020, MNRAS, 497, 626 / Balcels & Carter 1993, A&A, 279, 376 / Balcels et al, 2001, AJ, 122, 1758 / Bilek et al, 2022, A&A, 660, 28 / Costantin et al, 2021, ApJ, 913, 125 / Damjanov et al, 2011, ApJ, 739, L44 / Dekel et al, 2009, ApJ, 703, 785 / Dupraz & Combes, 1986, A&A, 166, 53 / Emsellem et al, 2004, MNRAS, 352, 721 / Falcon-Barroso et al, 2006, MNRAS 369, 529 / Falcon-Barroso & Martig, 2021, A&A, 646, A31 / Franx, et al, 2008, ApJ, 688, 770 / Frigo & Balcels 2017, MNRAS, 469, 2184 / Hau et al, 1996, IAU Symp 171, 388 / Kuijken & Merrifield 1993, 264, 712 / Malin & Carter 1983, ApJ, 274, 534 / Peralta de Arriba et al, 2015, MNRAS, 453, 704 / Perez-Gonzalez et al, 2008, ApJ, 675, 274 / Quinn 1984, ApJ, 279, 596 / Sanchez et al, 2023, MNRAS, subm, arXiv:2304.0122v1 / Sarzi et al, 2006, MNRAS, 366, 1151 / Sikkema et al, 2007, A&A, 467, 1011 / Toomre, A., 1977, EGSP Conf, 401 /

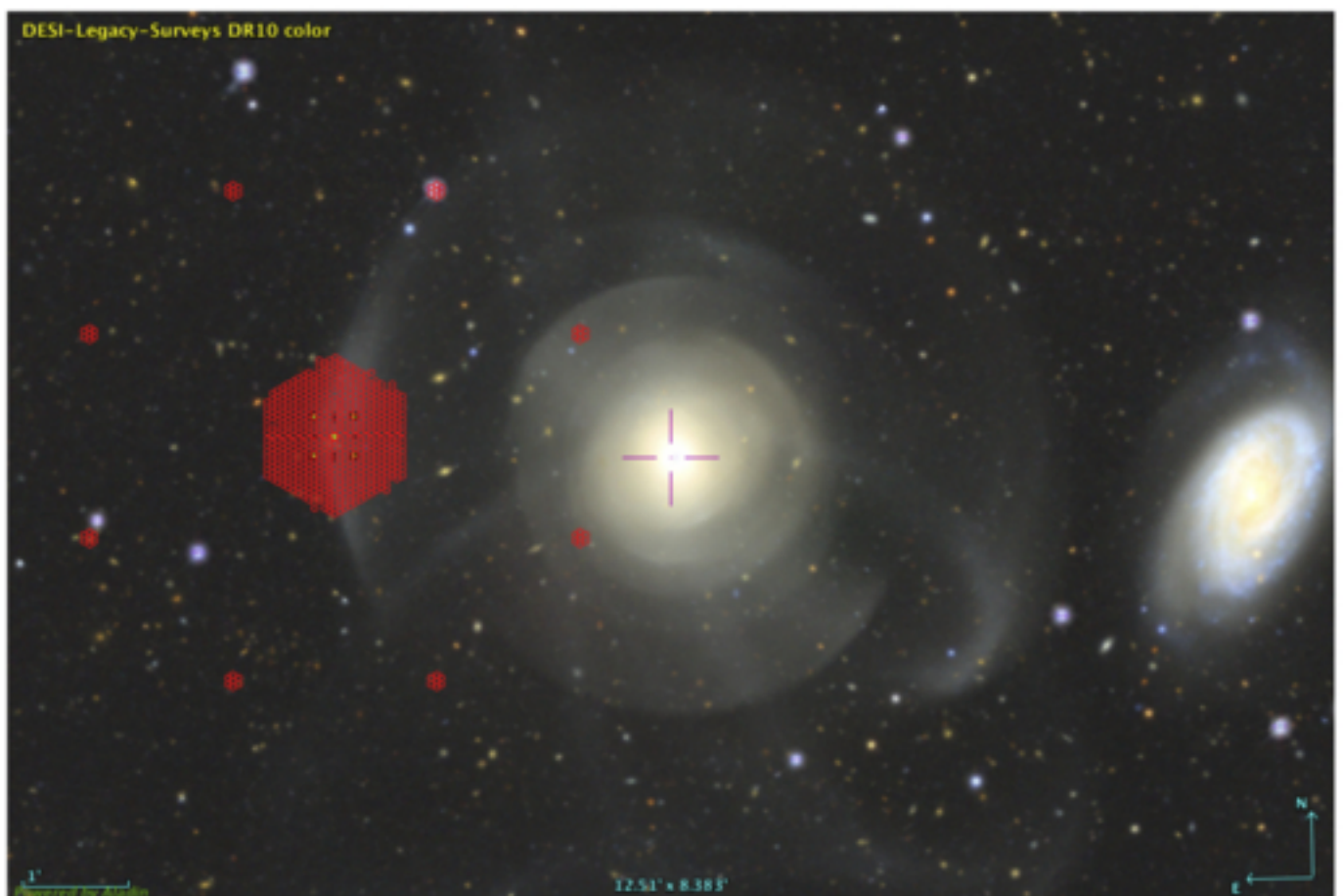
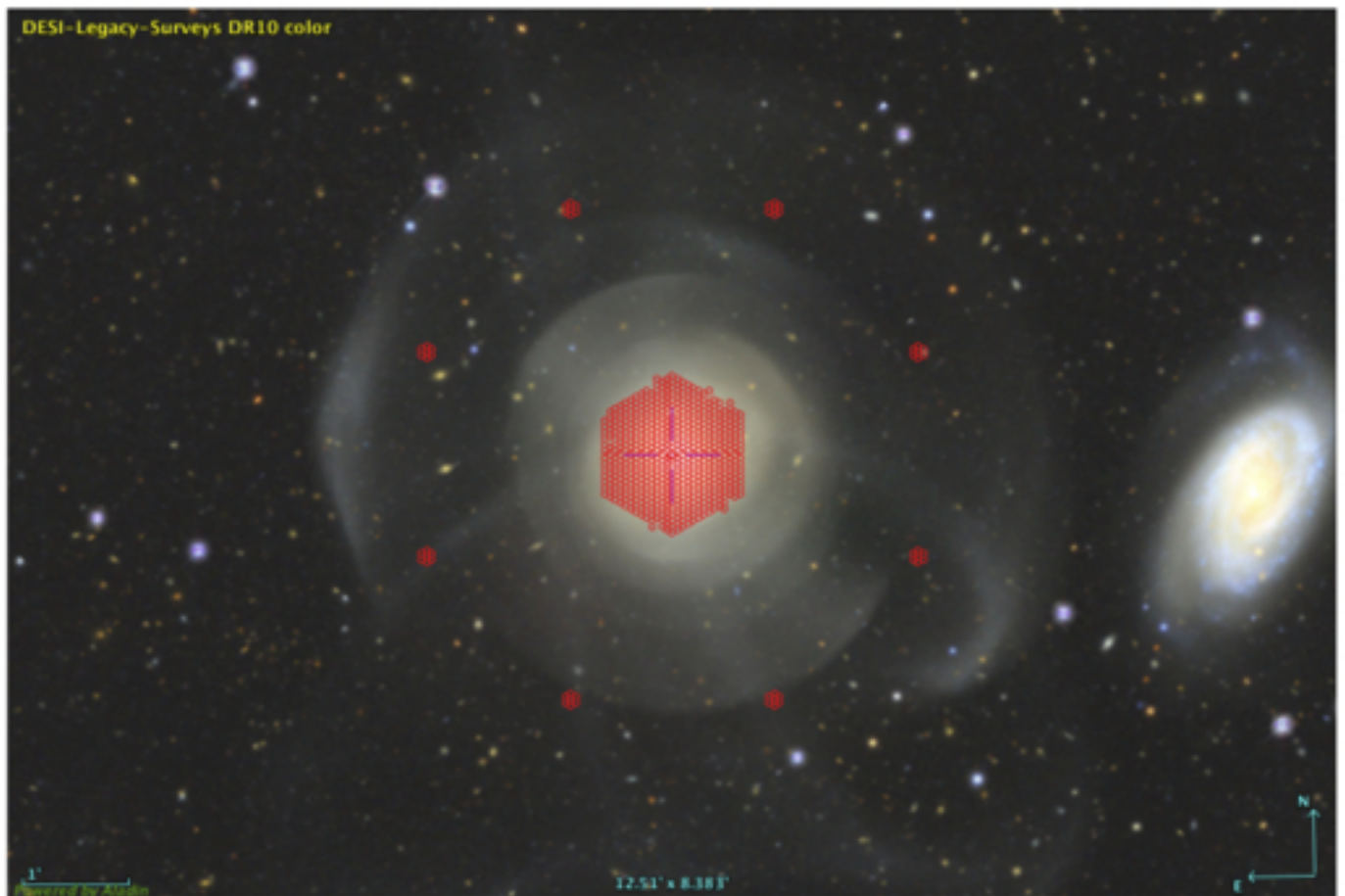


Figure 1: Chosen footprints of the WEAVE LIFU on the nucleus and East outer shell of NGC 474

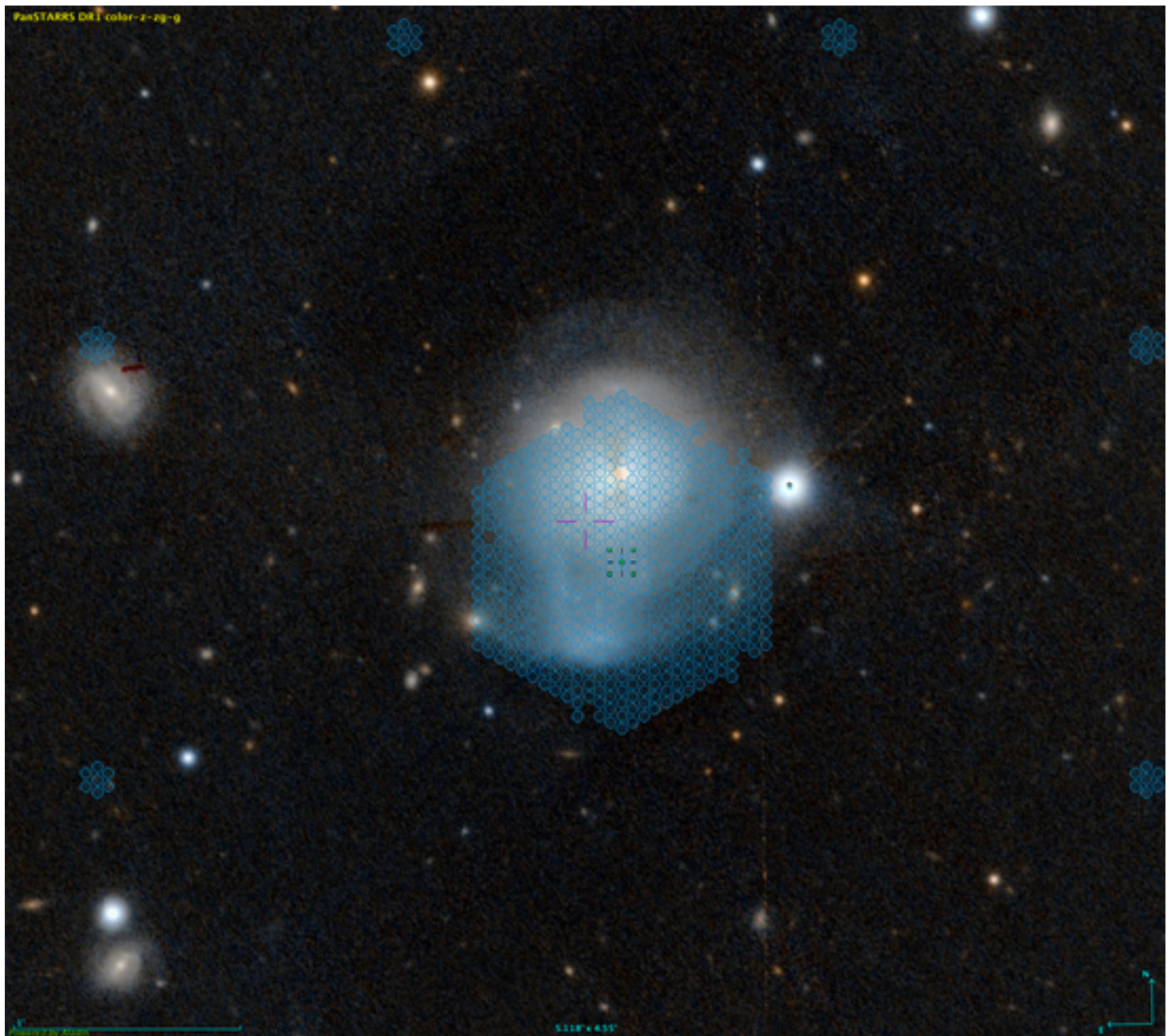


Figure 2: Chosen footprint of the WEAVE LIFU on NGC 3656, encompassing the centre and the southern bright shell.

Observing conditions, instrument setup and targets

Total time on targets hours

Sky brightness

Transparency

Worst FWHM arcsec

Maximum airmass

Focal-plane mode

Spectrograph mode

Spectral binning

Observing mode

Targets

NGC0474NUC 01:20:06.70 +03:24:55.0 J2000 V=11.51 LIFU+LR_1x1 6*1020
NGC0474NUC 01:20:06.70 +03:24:55.0 J2000 V=11.51 LIFU+HR_GR_1x1 6*1020
NGC0474SH 01:20:20.00 +03:25:08.1 J2000 SB=XX, LIFU+LR_2x2 6*1020
NGC3656 11:23:38.70 +53:50:31.9 J2000 V=12.4 LIFU+LR_2x2 6*1020
NGC5982 15:38:39.70 +59:21:20.0 J2000 V=11.3 LIFU+LR_1x1 6*1020
NGC5982 15:38:39.70 +59:21:20.0 J2000 V=11.3 LIFU+HR_GR_1x1 6*1020
NGC7626 23:20:42.29 +08:13:02.5 J2000 V=11.11 LIFU+LR_1x1 6*1020
NGC7626 23:20:42.29 +08:13:02.5 J2000 V=11.11 LIFU+HR_GR_1x1 6*1020